

Use a separate answer sheet for each problem. Make sure to write your name, student ID and the problem number on each sheet.

1. (a) Find the solution of $Ax = b$:
$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

(b) Find the inverses of $A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 7 & 0 & 0 \\ 5 & 6 & 0 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 4 & 3 \end{bmatrix}$.

2. Tridiagonal matrices have zero entries except on the main diagonal and the two adjacent diagonals. Factor these symmetric matrices into $A = LDL^T$.

(a) $A = \begin{bmatrix} c & c & 0 \\ c & c+b & b \\ 0 & b & b+a \end{bmatrix},$

(b) $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$

3. Suppose $Q^T = Q^{-1}$ (transpose equals inverse, so $Q^T Q = I$).

(a) Show that the columns $q_1, q_2, \dots, q_n$ are unit vectors: $\|q_i\|^2 = 1$.

(b) Show that every two columns of Q are perpendicular: $q_i^T q_j = 0$ if $i \neq j$.

(c) Find a 2 by 2 example with first entry $q_{21} = \sin\theta$

4. There are three lines on the xy plane

$$\begin{array}{ll} L1: & 2x + 2y = 2 \\ L2: & 2x + y = 2 \\ L3: & x + 2y = 2 \end{array}$$

(a) Find the solution of $L1$ and $L2$ as $\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

(b) Find the point $P(x_0, y_0)$, which minimize the function $f(x, y) = (2x + 2y - 2)^2 + (2x + y - 2)^2 + (x + 2y - 2)^2$.
(hint: $A^T A x = A^T b$)